



# The Papyrus Handbook

A guide to Markdown books with Papyrus

milon/papyrus

# The Papyrus Handbook

Markdown to PDF, EPUB, HTML sites, and KDP

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# Welcome

Thanks for opening **The Papyrus Handbook**. Use the sidebar to browse chapters, or continue from Introduction. The home page has the project banner, a **Downloads** link for PDF previews from GitHub, and links to GitHub and Packagist.

This guide is the companion for [milon/papyrus](https://github.com/milon/papyrus) — a PHP CLI that turns Markdown into PDF, EPUB, HTML, a multi-page site, and Amazon KDP exports. You are reading those same chapters as a Papyrus site.



# How to use this guide

1. **Introduction** — what Papyrus is and the project layout
2. **Install** — Composer setup and `papyrus init`
3. **Configuration** — every `papyrus.php` option
4. **Writing** — Markdown, front matter (`pretoc`, `draft`), asides, Mermaid, linting
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6. **build** — all-in-one export, watch, caches
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# Introduction

Papyrus is a Composer package and CLI for authors who write books in Markdown and need more than one delivery format from a single tree of chapters.

One project layout yields:

- **PDF** — print interiors and shareable digital books
- **EPUB** — stores, e-readers, and Kindle upload
- **HTML** — a single file with light / dark mode
- **Site** — a multi-page static website with popup search (this handbook)
- **KDP helpers** — Kindle EPUB, print PDF, covers, wrap estimates, metadata, package zip

This handbook is itself a Papyrus book. The Markdown source lives in [examples/the-papyrus-handbook/](https://github.com/ericniebler/examples/tree/master/the-papyrus-handbook) on GitHub. Prebuilt PDF, HTML, and site outputs are published from that example so you can read without installing anything.

## Who it is for

- Authors shipping a technical or long-form book from Markdown
- Teams migrating off [ibis-next](#)
- Anyone who wants PDF and a hosted docs site from the same chapters

## Project shape

Every book is a directory with:

| Path                     | Role                                                                                       |
|--------------------------|--------------------------------------------------------------------------------------------|
| <code>papyrus.php</code> | Title, trim, themes, Mermaid, samples, KDP                                                 |
| <code>content/</code>    | Markdown chapters (natural filename order)                                                 |
| <code>assets/</code>     | Your overrides (covers, banner, custom themes/fonts);<br>bundled defaults used when absent |
| <code>export/</code>     | Build outputs (usually gitignored)                                                         |

## Next steps

Install Papyrus (next chapter), run `papyrus doctor`, then `papyrus build` or `papyrus build:site`. Later chapters document every build and KDP command and the full `papyrus.php` option set.

Source and releases: [github.com/milon/papyrus](https://github.com/milon/papyrus).

# Install and project layout

# Requirements

## Required

- PHP 8.2+ with extensions `dom`, `gd`, `mbstring`, `zip`, and `zlib`
- Composer

## Optional

| Tool                                                                                            | Used by                | Notes                                                |
|-------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------|
| <code>mmdc</code> ( <code>@mermaid-js/mermaid-cli</code> or Homebrew <code>mermaid-cli</code> ) | Mermaid diagrams       | Needs Chrome/Chromium for Puppeteer                  |
| Chrome or Chromium                                                                              | Mermaid CLI            | Set <code>PUPPETEER_EXECUTABLE_PATH</code> if needed |
| <code>epubcheck</code>                                                                          | <code>kdp:ebook</code> | Extra validation; skipped with a warning when absent |

## macOS (Homebrew)

```
brew install php composer
brew install mermaid-cli
brew install --cask google-chrome
brew install epubcheck
```

```
export PUPPETEER_EXECUTABLE_PATH="/Applications/Google  
Chrome.app/Contents/MacOS/Google Chrome"
```

## npm / Linux

```
npm install -g @mermaid-js/mermaid-cli  
# Debian/Ubuntu example:  
sudo apt-get install -y chromium-browser  
export PUPPETEER_EXECUTABLE_PATH="$(command -v chromium-browser  
|| command -v google-chrome || command -v chromium)"
```

Install [epubcheck](#) and put it on **PATH**, or use **brew install epubcheck** on macOS.

## Check the toolchain

```
php -m | grep -E 'dom|gd|mbstring|zip|zlib'  
mmdc --version  
epubcheck --version  
papyrus doctor
```



## Install per project

In your book repository:

```
composer require milon/papyrus
```

```
vendor/bin/papyrus --version  
vendor/bin/papyrus init  
vendor/bin/papyrus doctor
```

Wire Composer scripts (example):

```
{  
  "scripts": {  
    "build": "papyrus build",  
    "build:pdf": "papyrus build:pdf --theme light,dark",  
    "build:epub": "papyrus build:epub",  
    "build:html": "papyrus build:html",  
    "build:site": "papyrus build:site",  
    "build:sample": "papyrus build:sample",  
    "build:kdp": "papyrus kdp",  
    "doctor": "papyrus doctor"  
  }  
}
```

Package page: [packagist.org/packages/milon/papyrus](https://packagist.org/packages/milon/papyrus).

## Install globally

```
composer global require milon/papyrus
export PATH="$(composer global config bin-dir --absolute):$PATH"
papyrus list
```

## Scaffold a book

```
papyrus init  
papyrus init -d my-book
```

`init` creates `papyrus.php`, `content/`, and an empty `assets/` directory. Papyrus uses bundled themes, CSS, and fonts by default. Publish those files into your project only when you want to customize them:

```
papyrus asset:publish  
papyrus asset:publish --only=themes,css
```

Use `--force` / `-f` to overwrite files during `init`, or with `asset:publish` to overwrite published assets. `--only` limits publishing to `themes`, `css`, and/or `fonts`.

## Layout

| Path                     | Role                                               |
|--------------------------|----------------------------------------------------|
| <code>papyrus.php</code> | Book settings                                      |
| <code>content/</code>    | Markdown chapters                                  |
| <code>assets/</code>     | Your overrides: themes, CSS, covers, fonts, banner |
| <code>export/</code>     | Built artifacts                                    |
| <code>.papyrus/</code>   | Incremental caches                                 |

Always run from the book root, or pass `-d / --dir`. Override where artifacts are written with `-e / --export` (default: `<book>/export`):

```
papyrus doctor -d /path/to/book
papyrus build --dir /path/to/book
papyrus build:site -d /path/to/book -e /path/to/out
```

Browse the Papyrus source on [GitHub](#) if you want to follow along with this handbook's own project under `examples/the-papyrus-handbook/`.

# Configuration

All settings live in `papyrus.php`, which returns a PHP array.

## Identity and paths

```
return [  
    'title' ⇒ 'My Book',  
    'subtitle' ⇒ 'A short subtitle',  
    'author' ⇒ 'Author Name',  
    'language' ⇒ 'en',  
    'themes' ⇒ ['light', 'dark'],  
  
    'content_dir' ⇒ 'content',  
    'assets_dir' ⇒ 'assets',  
    'export_dir' ⇒ 'export',  
];
```

| Key                      | Default                        | Notes                                            |
|--------------------------|--------------------------------|--------------------------------------------------|
| <code>title</code>       | <code>Untitled</code>          | Drives the export filename slug                  |
| <code>subtitle</code>    | <code>''</code>                |                                                  |
| <code>author</code>      | <code>''</code>                |                                                  |
| <code>language</code>    | <code>en</code>                | EPUB / KDP metadata fallback                     |
| <code>themes</code>      | <code>['light', 'dark']</code> | Each needs <code>assets/theme-{name}.html</code> |
| <code>content_dir</code> | <code>content</code>           | Relative to the book root                        |

| Key                     | Default             | Notes                                                 |
|-------------------------|---------------------|-------------------------------------------------------|
| <code>assets_dir</code> | <code>assets</code> |                                                       |
| <code>export_dir</code> | <code>export</code> | Overridden by <code>-e</code> / <code>--export</code> |

The slug is the title lowercased with non-alphanumeric runs turned into `-`. This handbook is *The Papyrus Handbook*, so outputs look like `the-papyrus-handbook-light.pdf`.

## Page size and margins

```
'document' => [  
  'size' => 'crown-quarto',  
  'margin_left' => 27,  
  'margin_right' => 27,  
  'margin_top' => 14,  
  'margin_bottom' => 14,  
],
```

List presets with **papyrus sizes**. Custom trim:

```
'document' => [  
  'format' => [152.4, 228.6], // width_mm, height_mm  
],
```



## Table of contents

```
'toc' => [  
  'h1' => 0,  
  'h2' => 0,  
  'h3' => 1,  
],
```

Values follow mPDF **h2toc** conventions used by Papyrus (**0** include, **1** exclude).

## Cover images

```
'cover' => [  
  'image' => 'cover.png',  
  'light' => 'cover-light.png',  
  'dark' => 'cover-dark.png',  
  // 'width' => 188.976, // mm; defaults to page width  
  // 'height' => 246.126, // mm; defaults to page height  
],
```

Paths are relative to **assets/**. Missing per-theme covers fall back to **cover.image**. KDP print PDFs omit the cover page entirely.

## Running header

```
'header' => [  
  'style' => 'font-style: italic; text-align: right; border-  
bottom: solid 1px #808080;',  
],
```

Pretoc chapters suppress folios differently from body chapters.

## Fonts

Declare faces under `assets/fonts/` and optional **script routing** (PDF only). First matching **script** rule wins.

```
'fonts' => [
  'default' => 'librelibertine', // optional; else first face,
  else mPDF default
  'faces' => [
    [
      'name' => 'librelibertine',
      'regular' => 'LinLibertine_R.ttf',
      'bold' => 'LinLibertine_RB.ttf',
      'italic' => 'LinLibertine_RI.ttf',
      'bold_italic' => 'LinLibertine_RBI.ttf',
    ],
    [
      'name' => 'notosansbengali',
      'regular' => 'NotoSansBengali-Regular.ttf',
      'otl' => true, // OpenType layout – needed for
complex scripts
    ],
  ],
  'script' => [
    // When mPDF tags a run as Bengali, use this face instead
of the default
    ['match' => ['bn', 'ben', 'bengali'], 'face' =>
'notosansbengali'],
  ],
],
```

## Faces

Each face needs a **name** and a **regular** file that exists under **assets/fonts/**. Optional keys: **bold**, **italic**, **bold\_italic**, and **otl** (sets mPDF **useOTL** when true — common for code fonts and Indic / Arabic scripts).

Papyrus merges these into mPDF's **fontdata** when building PDF (**FontRegistry** → **MpdfFactory**).

## Script rules

**fonts.script** is a list of { **match**, **face** } rules used only for **PDF**. They do not change EPUB or HTML typography.

When any applicable rule exists, Papyrus turns on mPDF's **autoScriptToLang** and **autoLangToFont**, and installs a custom **languageToFont** resolver (**ScriptLanguageToFont**). mPDF then:

1. Detects script runs in the HTML (e.g. Bengali glyphs).
2. Tags them with a language / script code (e.g. **bn**, or **\*-beng**).
3. Asks Papyrus which face to use; the first rule whose **match** list hits that code wins.

**match** aliases are lowercased. Built-in aliases include:

| Aliases you can list                                  | Resolves as |
|-------------------------------------------------------|-------------|
| <b>bn</b> , <b>ben</b> , <b>beng</b> , <b>bengali</b> | Bengali     |

| Aliases you can list                          | Resolves as       |
|-----------------------------------------------|-------------------|
| <code>ar, arab, arabic</code>                 | Arabic            |
| <code>hi, hin, deva, devanagari, hindi</code> | Devanagari        |
| any 4-letter ISO script code                  | that script as-is |

The `face` string must be a name from `fonts.faces` and that face must actually load (regular file present). Rules pointing at a missing face are skipped (`FontRegistry::applicableScriptRules()`). Unmatched languages fall through to mPDF's built-in `LanguageToFont`.

Example: body text in Latin uses `librelibertine`; a paragraph containing বন্ধন is auto-tagged Bengali and rendered with `notosansbengali` if that face is registered.

**Notice:** Without both a registered face and a matching `script` rule, non-Latin runs often show as tofu (missing glyphs) in the default font.

## Site home (**build:site**)

```
'site' => [
  'banner' => 'banner.jpg',
  'lead' => 'A one-line pitch for the home page.',
  'cname' => 'docs.example.com',
  'base_path' => '/my-repo', // project Pages path; omit for
  custom domains at /
  'links' => [
    ['label' => 'Downloads', 'chapter' => '19-downloads.md'],
    ['label' => 'Source on GitHub', 'url' =>
'https://github.com/you/your-book'],
    ['label' => 'Issues', 'url' =>
'https://github.com/you/your-book/issues'],
  ],
],
```

| Key           | Default                                                     | Notes                                                                                                                                             |
|---------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>banner</b> | auto <b>banner.jpg</b><br>/ <b>banner.png</b> if<br>present | Under <b>assets/</b>                                                                                                                              |
| <b>lead</b>   | unset                                                       | Home page pitch                                                                                                                                   |
| <b>cname</b>  | unset                                                       | Custom domain; writes a <b>CNAME</b><br>file in the site root for GitHub<br>Pages; absolute <b>sitemap.xml</b> /<br><b>robots.txt</b> Sitemap URL |

| Key                    | Default           | Notes                                                                                                                                   |
|------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>base_path</code> | unset (site at /) | Path prefix for project GitHub Pages (e.g. <code>/my-repo</code> ); injects <code>&lt;base href&gt;</code> ; also prefixes sitemap locs |
| <code>links</code>     | unset             | Explicit home-page links; each item needs <code>label</code> plus either <code>url</code> or <code>chapter</code>                       |

`chapter` matches a chapter source name like `19-downloads.md`, `19-downloads`, or a full relative source path, and links to that generated page. `repository` still works as a legacy fallback that auto-adds GitHub, Packagist, and Issues links when `links` is not set. Chapters are never linked automatically; add them explicitly through `links`.



## Mermaid

```
'mermaid' => [  
  'enabled' => true,  
  'format' => 'svg',           // svg | png  
  'theme' => 'auto',          // auto = book colours; or  
                             default/dark/forest/neutral  
  'max_width_mm' => 130,  
  // 'command' => '/usr/local/bin/mmdc', // optional override  
],
```

| Key          | Default                     | Notes                                                       |
|--------------|-----------------------------|-------------------------------------------------------------|
| enabled      | false                       | Requires <b>mmdc</b> on <b>PATH</b> (or <b>command</b> )    |
| format       | svg                         | Anything other than <b>png</b> becomes <b>svg</b>           |
| theme        | auto                        | Book-matched palettes for PDF; HTML/site embed light + dark |
| max_width_mm | 130                         | PDF figure width cap                                        |
| command      | auto-resolve<br><b>mmdc</b> | Explicit CLI path                                           |

## Sample PDF

```
'sample' => [  
  'ranges' => [  
    ['from' => 1, 'to' => 3],  
  ],  
  'chapters' => [  
    '01-introduction.md',  
  ],  
],  
'sample_notice' => 'This is a sample from My Book.'
```

Ranges are 1-based inclusive pages of the finished theme PDF. **chapters** lists content filenames (or stems) to include in full. Notice text also accepts **sample.notice** / **sample.text** as fallbacks. See the **build:sample** chapter.

# KDP

```
'kdp' => [
  'ebook' => [
    'enabled' => true,
    'cover' => 'cover-ebook.jpg',
  ],
  'print' => [
    'enabled' => true,
    'bleed_mm' => 3,
    'margin_preset' => 'recommended', // or minimum
    'paper' => 'white',                // or cream - spine
    width
    'back_cover' => 'back-cover.png', // optional wrap
    artwork
    'spine_color' => '#1a1a1a',        // optional wrap spine
    fill
  ],
  'metadata' => [
    'description' => 'Bookstore blurb...',
    'keywords' => ['laravel', 'filament'],
    'language' => 'en',
  ],
],
```

Full option tables live in the **kdp** and **kdp:\*** chapters. **papyrus build** only runs KDP when ebook or print is enabled; **papyrus kdp** errors if neither is.

## Break level

```
'break_level' => 2,
```

Automatic page breaks before headings (**1** = H1, **2** = H1 and H2). Explicit **[break]** markers always insert a break.

## CommonMark hook

```
'configure_commonmark' => function  
(\League\CommonMark\Environment\Environment $environment): void {  
    // register custom extensions  
},
```

# Writing content

Chapters are Markdown under `content/`, loaded in natural filename order (`00-...`, `01-...`, `10-...`).

## Front matter

```
---
title: My chapter
pretoc: true
---

Body starts here.
```

| Key                 | Meaning                                                                                |
|---------------------|----------------------------------------------------------------------------------------|
| <code>title</code>  | Chapter title for EPUB / site sidebar / running headers                                |
| <code>pretoc</code> | <code>true</code> places the chapter <b>before</b> the PDF table of contents           |
| <code>draft</code>  | <code>true</code> omits the chapter from PDF, EPUB, HTML, site, sample, and KDP builds |

Accepted truthy values for `pretoc` and `draft`: `true`, `"true"`, `1`, `"1"`.

Pass `--include-drafts` on any book build command (or `watch`) when you want draft chapters in the output. `doctor` lists drafts and warns if `site.links` points at one.

## Pretoc chapters

In the PDF, Papyrus splits the book into three bands around the theme's TOC marker (`<!-- PAPYRUS:TOC -->`):

1. **Pretoc** chapters (`pretoc: true`) — title page material, copyright, dedication, welcome, etc.
2. **Table of contents** — generated from body headings (see `toc` in config)
3. **Body** chapters — everything else, with folios and the running header

Pretoc chapters:

- Appear in EPUB, HTML, and the site like any other chapter
- Do **not** get normal body folios the same way (front matter styling)
- Are omitted from the auto-generated PDF TOC as body entries
- Still sort by filename — use a low prefix such as `00-` so they come first

This handbook's Welcome page is the example:

```
---
title: Welcome
pretoc: true
---

# Welcome

...
```

File: `content/00-welcome.md`. Introduction and later chapters omit



`pretoc` (or set it false) so they sit after the TOC in the PDF.

You can have several pretoc files (`00-copyright.md`, `00-dedication.md`, `00-welcome.md`); all of them render before the TOC, in filename order.

# Markdown features

CommonMark + GFM, fenced code highlighting, and book extras.

## Emphasis and code

```
Hello, **world**.
```

```
`inline code`
```

```
```php
$user = User::factory()→create();
```
```

## Callouts

```
> {notice} Something helpful.
```

```
> {warning} Something risky.
```

```
 ::: note
A note aside.
 :::

 ::: warning
A warning aside.
 :::
```

## Page breaks

```
[break]
```

## Mermaid

Write a **mermaid** fence in your chapter. Papyrus runs **mmdc** at build time and replaces the fence with an SVG (or PNG) figure in PDF, EPUB, HTML, and site exports.

Source:

```

```mermaid
flowchart TB
    subgraph Authoring
        MD[Markdown chapters]
        CFG[papyrus.php]
        ASSETS[Themes and fonts]
    end

    subgraph Papyrus
        CONVERT[BookConverter]
        MERMAID[MermaidRenderer]
        PDF[build:pdf]
        EPUB[build:epub]
        HTML[build:html]
        SITE[build:site]
    end

    MD --> CONVERT
    CFG --> CONVERT
    ASSETS --> PDF
    ASSETS --> HTML
    ASSETS --> SITE
    CONVERT --> MERMAID
    MERMAID --> PDF
    MERMAID --> EPUB
    MERMAID --> HTML
    MERMAID --> SITE
```

```

Rendered output:



Enable in `papyrus.php` (theme defaults to `auto` — book colours; HTML and site embeds both light and dark variants):

```
'mermaid' => [  
    'enabled' => true,  
],
```

Optional knobs: `format` (`svg` / `png`), `theme` (`auto` or a Mermaid stock theme like `default` / `dark` / `forest`), `max_width_mm`.

Requires `@mermaid-js/mermaid-cli` (`mmdc`) on `PATH`. Diagrams cache under `.papyrus/cache/mermaid`.

## PHP fence linting

```
papyrus lint
papyrus lint --fix
papyrus lint --max-width=66
```

| Option                        | Short                | Default         | Meaning                                                      |
|-------------------------------|----------------------|-----------------|--------------------------------------------------------------|
| <code>--fix</code>            | <code>-f</code>      | off             | Apply auto-fixes for open tags and comment runs              |
| <code>--max-width</code>      | —                    | <code>66</code> | Maximum line width for PHP fences                            |
| <code>--dir / --export</code> | <code>-d / -e</code> | book defaults   | Shared book options ( <code>--export</code> unused for lint) |

# Themes and assets

Everything visual can live under **assets/**, but it does not have to. New projects from **papyrus init** start with an empty **assets/** directory and use the bundled Papyrus defaults: Linux Libertine for body and headings, 0xProto for code, github-gist highlight colors, notice / tip / caution asides, and matching font files.

Publish the bundled assets into your project when you want to edit them:

```
papyrus asset:publish
papyrus asset:publish --only=themes,css
papyrus asset:publish --only=fonts --force
```

**--only** accepts **themes**, **css**, and/or **fonts**. Omit it to publish everything.

## PDF themes

For each name in **themes**, Papyrus looks for **assets/theme-{name}.html** first, then falls back to the bundled copy. Split the theme on the TOC marker:

```
<!-- PAPYRUS:TOC -->
```

Papyrus writes pretoc chapters, then the TOC, then body chapters around that marker. A legacy **<!-- IBIS:TOC -->** marker is still accepted. **migrate-ibis** rewrites that marker only in **your assets/theme\*.html** files, not in the bundled defaults.

Typical theme responsibilities: **@page** / typography, code blocks, callout colors, title page, and running header styles (often paired with **header.style** in config).



## HTML theme

`theme-html.html` is a full HTML document with placeholders `{{ $title }}`, `{{ $subtitle }}`, `{{ $author }}`, and `{{ $body }}`. The default template mirrors the light PDF theme (same typefaces, colors, code blocks, and callouts), with a fixed moon/sun icon toggle that switches to the dark PDF palette. The reading column follows Tailwind screen widths (640 → 1536px). `@font-face` rules in the theme use `../assets/fonts/`. For `build:html`, Papyrus embeds those font files as base64 data URIs so the export is self-contained. For `build:site`, fonts are copied into the site `assets/fonts/` folder.

```
papyrus build:html
papyrus build:site
```

`build:site` writes `export/<slug>-site/` with one HTML page per chapter, a Home index, shared CSS/JS (fonts copied into the site), a collapsible chapter sidebar, and the same light/dark toggle as the single-file HTML export. That folder is what this handbook deploys to GitHub Pages.

## EPUB styles

`style.css` (and optional `highlight.codeblock.min.css`) are loaded from your project `assets/` first, then from Papyrus's bundled defaults, and ship inside the EPUB. Prefer simple `pre/code` rules for e-ink readers.

## Covers and fonts

Place cover images and font files under `assets/` (fonts usually in `assets/fonts/`). Reference them from `cover` and `fonts.faces` in `papyrus.php`.

This handbook keeps only book-specific assets in `assets/` — `cover.jpg` for the PDF cover and `banner.jpg` for the site home page. Themes, CSS, and fonts come from Papyrus's bundled defaults; run `papyrus asset:publish` when you want local copies to customize. The wide banner image in the Papyrus repo README lives at [assets/papyrus-banner.jpg](#).

# build

**papyrus build** is the all-in-one export command. From the book root (or with **-d**), it runs every configured PDF theme, then EPUB, then single-file HTML, then every **enabled** KDP output. Site and sample builds are opt-in.

```
papyrus build
papyrus build --with-site --with-sample
papyrus build -d /path/to/book
papyrus build -d examples/the-papyrus-handbook -e docs --with-site --with-sample
```

## Options

| Option                        | Short           | Default                                                     | Meaning                                            |
|-------------------------------|-----------------|-------------------------------------------------------------|----------------------------------------------------|
| <code>--dir</code>            | <code>-d</code> | current directory                                           | Book root (must contain <code>papyrus.php</code> ) |
| <code>--export</code>         | <code>-e</code> | <code>export_dir</code> from config ( <code>export</code> ) | Where artifacts are written                        |
| <code>--with-site</code>      |                 | off                                                         | Also run <code>build:site</code>                   |
| <code>--with-sample</code>    |                 | off                                                         | Also run <code>build:sample</code> for every theme |
| <code>--include-drafts</code> |                 | off                                                         | Include chapters with <code>draft: true</code>     |

Global Symfony options also apply (`-v` / `--verbose`, `-q`, `--no-ansi`, ...).  
 With `-v`, Papyrus prints third-party notices from mPDF and the EPUB zipper.

## What it builds

| Step                                              | Always?                                                                           | Output                                                    |
|---------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------|
| PDF<br>(each <b>themes</b> entry)                 | yes                                                                               | <code>export/&lt;slug&gt;-&lt;theme&gt;.pdf</code>        |
| EPUB                                              | yes                                                                               | <code>export/&lt;slug&gt;.epub</code>                     |
| HTML                                              | yes                                                                               | <code>export/&lt;slug&gt;.html</code>                     |
| KDP<br>ebook /<br>print /<br>covers /<br>metadata | only if<br><code>kdp.ebook.enabled</code><br>or<br><code>kdp.print.enabled</code> | see KDP chapters                                          |
| Site                                              | only with <code>--with-site</code>                                                | <code>export/&lt;slug&gt;-site/</code>                    |
| Sample<br>PDF                                     | only with <code>--with-sample</code>                                              | <code>export/sample-&lt;slug&gt;-&lt;theme&gt;.pdf</code> |

Slug comes from **title** (lowercased, non-alphanumeric → -).

**watch** invokes **build** on each change. Pass `--with-site` and/or `--with-sample` to include those outputs in every rebuild. Pass `--include-drafts` to keep **draft: true** chapters in the rebuild.

## Failures

Each step is independent. If one format fails, Papyrus continues with the others and exits non-zero if any step failed. When no KDP outputs are enabled, the KDP block is skipped silently (unlike `papyrus kdp`, which errors).

## Validate first

```
papyrus doctor  
papyrus build
```



## Watch mode

Rebuild on changes to `papyrus.php`, `content/`, and `assets/`:

```
papyrus watch
papyrus watch --interval 3
papyrus watch --with-site --with-sample
```

| Option                        | Short                | Default                    | Meaning                              |
|-------------------------------|----------------------|----------------------------|--------------------------------------|
| <code>--interval</code>       | <code>-i</code>      | 2                          | Poll interval in seconds (minimum 1) |
| <code>--dir / --export</code> | <code>-d / -e</code> | same as <code>build</code> | Passed through to each rebuild       |
| <code>--with-site</code>      |                      | off                        | Also run <code>build:site</code>     |
| <code>--with-sample</code>    |                      | off                        | Also run <code>build:sample</code>   |

## Caching

Chapter HTML caches under `.papyrus/cache/markdown`. Mermaid figures use `.papyrus/cache/mermaid`. Delete `.papyrus/` for a cold rebuild.

## Related chapters

The next chapters cover each `build:*` command and every KDP command in turn.

# build:pdf

Build one PDF per theme (or a subset). Each theme needs `assets/theme-{name}.html` and an entry in `themes`.

```
papyrus build:pdf
papyrus build:pdf --theme light
papyrus build:pdf --theme light,dark --parallel
papyrus build:pdf -d /path/to/book -e /path/to/out
```

## Options

| Option                  | Short           | Default               | Meaning                     |
|-------------------------|-----------------|-----------------------|-----------------------------|
| <code>--dir</code>      | <code>-d</code> | current directory     | Book root                   |
| <code>--export</code>   | <code>-e</code> | <code>export/</code>  | Output directory            |
| <code>--theme</code>    | <code>-t</code> | all configured themes | Comma-separated theme names |
| <code>--parallel</code> | <code>-p</code> | off                   | One process per theme       |

## Output

```
export/<slug>-<theme>.pdf
```

Example for this handbook: `the-papyrus-handbook-light.pdf`.

## Config that affects PDF

| Key                                          | Role                                      |
|----------------------------------------------|-------------------------------------------|
| <code>themes</code>                          | Which theme names to build by default     |
| <code>document.size / document.format</code> | Trim size                                 |
| <code>document.margin_*</code>               | Margins in millimetres                    |
| <code>cover</code>                           | Cover image(s) under <code>assets/</code> |
| <code>header.style</code>                    | Running header CSS                        |
| <code>toc</code>                             | Which heading levels enter the TOC        |
| <code>break_level</code>                     | Auto page breaks before headings          |
| <code>fonts</code>                           | Faces and script routing                  |
| <code>mermaid</code>                         | Diagram rendering                         |

List trim presets with `papyrus sizes`. Custom trim:

```
'document' => [  
  'format' => [152.4, 228.6], // width_mm, height_mm  
],
```



## Parallel builds

`--parallel` / `-p` spawns one PHP process per theme. Useful when `themes` has both `light` and `dark`. Sequential is the default (and what `papyrus build` uses).

## Covers

```
'cover' => [  
  'image' => 'cover.png',          // fallback for every theme  
  'light' => 'cover-light.png',    // optional per-theme  
  'dark' => 'cover-dark.png',  
  // 'width' / 'height' - mm; default to page size  
],
```

Paths are relative to **assets/**. Missing per-theme covers fall back to **cover.image**.

# build:epub

Build an EPUB3 package from the same Markdown chapters as PDF and HTML.

```
papyrus build:epub  
papyrus build:epub -d /path/to/book  
papyrus build:epub -e /path/to/out
```

## Options

| Option                | Short           | Default              | Meaning          |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code>    | <code>-d</code> | current directory    | Book root        |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

## Output

```
export/<slug>.epub
```

## What goes in the package

- Chapter HTML derived from `content/`
- `assets/style.css` and optional `highlight.codeblock.min.css`
- Embedded images referenced from chapters
- Cover from `cover` (theme fallback → `cover.image`)
- Metadata from `title`, `subtitle`, `author`, and `language`

## Config that affects EPUB

| Key                                                                 | Role                                       |
|---------------------------------------------------------------------|--------------------------------------------|
| <code>title</code> , <code>subtitle</code> ,<br><code>author</code> | Package identity                           |
| <code>language</code>                                               | EPUB language (default <code>en</code> )   |
| <code>cover</code>                                                  | Cover image under <code>assets/</code>     |
| <code>mermaid</code>                                                | Pre-rendered figures                       |
| <code>fonts</code>                                                  | Faces are PDF-oriented; EPUB relies on CSS |

Prefer simple `pre` / `code` rules in `assets/style.css` for e-ink readers.

## Kindle upload

For Amazon KDP's Kindle pipeline, prefer `kdp:ebook` (validates and names the file for upload). `build:epub` is the general-purpose store / archive build.



# build:html

Build a **single-file** HTML book with light / dark mode.

```
papyrus build:html  
papyrus build:html -d /path/to/book -e docs
```

## Options

| Option                | Short           | Default              | Meaning          |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code>    | <code>-d</code> | current directory    | Book root        |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

## Output

```
export/<slug>.html
```

Requires `theme-html.html` (project `assets/` or bundled default).

Placeholders in the theme:

| Placeholder                   | Source                |
|-------------------------------|-----------------------|
| <code>{{ \$title }}</code>    | <code>title</code>    |
| <code>{{ \$subtitle }}</code> | <code>subtitle</code> |
| <code>{{ \$author }}</code>   | <code>author</code>   |
| <code>{{ \$body }}</code>     | Rendered chapters     |

The default stub theme mirrors the light PDF palette, with a moon/sun toggle that switches to the dark PDF colours. The reading column follows Tailwind screen widths (640 → 1536px).

`@font-face` rules that point at `../assets/fonts/` are embedded as base64 data URIs at build time, so the single HTML file is self-contained even when fonts come from Papyrus's bundled defaults.

## Config that affects HTML

| Key                                                                 | Role                                                                                 |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <code>title</code> , <code>subtitle</code> ,<br><code>author</code> | Document chrome                                                                      |
| <code>mermaid</code>                                                | Figures; <code>theme</code> $\Rightarrow$ <code>auto</code> embeds light + dark SVGs |
| <code>fonts</code>                                                  | Loaded via the HTML theme's <code>@font-face</code> rules                            |

## Site vs HTML

| Command                 | Result                                              |
|-------------------------|-----------------------------------------------------|
| <code>build:html</code> | One self-contained <code>.html</code> file          |
| <code>build:site</code> | Multi-page folder with sidebar, Home, shared CSS/JS |

Use `build:html` for emailing or archiving a single document; use `build:site` for hosting.

# build:site

Build a multi-page static site: Home index, one HTML page per chapter, sidebar navigation, light / dark mode, and Prev / Next links.

```
papyrus build:site
papyrus build:site -d examples/the-papyrus-handbook -e docs
```

**papyrus build** does **not** run this by default — pass **--with-site** on **build**, or call **build:site** directly.

Preview the site (including popup search) with PHP's built-in server:

```
papyrus serve
papyrus serve --build
papyrus serve --port 8080
papyrus serve -s docs/the-papyrus-handbook-site
papyrus serve -d examples/the-papyrus-handbook -e docs --build
```

By default **serve** reads **export/<slug>-site/**. Pass **--site** / **-s** to point at another folder (for example the handbook under **docs/**) — no book project is required in that case. **--export** / **-e** still changes the default parent (**<export>/<slug>-site**). Pass **--build** to run **build:site** first (into that same site directory; needs **-d** / a book root). If **site.base\_path** is set on a loaded project, the printed URL includes that prefix so **<base href>**



## Options

| Option                | Short           | Default              | Meaning                              |
|-----------------------|-----------------|----------------------|--------------------------------------|
| <code>--dir</code>    | <code>-d</code> | current directory    | Book root                            |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Parent directory for the site folder |



## Output

```
export/<slug>-site/  
  index.html  
  404.html  
  <chapter-slug>.html  
  sitemap.xml  
  robots.txt  
  .nojekyll  
  CNAME                # when site.cname is set  
  assets/site.css  
  assets/site.js  
  assets/search.json  
  assets/fonts/...  
  assets/<banner>      # when configured
```

Point any static host (GitHub Pages, Netlify, S3, ...) at that folder. `.nojekyll` tells GitHub Pages not to run Jekyll. `CNAME` (from `site.cname`) sets a custom domain. GitHub Pages and Netlify serve `404.html` for missing URLs. Press `/` or the topbar search button for popup search (`↑/↓` moves through results, Enter opens the hit, Escape closes). Results are ranked: title and heading matches outrank body hits, whole-word matches beat substrings, and pretoc chapters are demoted so Welcome/copyright noise sinks. Excerpts centre on the first matching term. `sitemap.xml` uses `https://{cname}` when `site.cname` is set, otherwise the `site.base_path` prefix. `robots.txt` points at the sitemap when a CNAME is configured. Each `##` heading gets an id and a `#` permalink; search hits for those headings open the chapter at that section.

## Site config

Optional **site** block in **papyrus.php**:

```
'site' => [
    'banner' => 'banner.jpg',           // under assets/; auto-
detects banner.jpg / banner.png
    'lead' => 'A one-line pitch for the home page.',
    'cname' => 'docs.example.com',      // GitHub Pages custom
domain
    'base_path' => '/my-repo',          // project Pages under
github.io/my-repo/; omit with cname
    'links' => [
        ['label' => 'Downloads', 'chapter' => '19-downloads.md'],
        ['label' => 'Source on GitHub', 'url' =>
'https://github.com/you/your-book'],
    ],
],
```

| Key           | Default                                                      | Meaning                                |
|---------------|--------------------------------------------------------------|----------------------------------------|
| <b>banner</b> | <b>banner.jpg</b><br>then<br><b>banner.png</b> if<br>present | Hero image on Home                     |
| <b>lead</b>   | unset                                                        | Short pitch under the title on<br>Home |

| Key                    | Default | Meaning                                                                                                  |
|------------------------|---------|----------------------------------------------------------------------------------------------------------|
| <code>cname</code>     | unset   | Writes <code>CNAME</code> in the site root for a GitHub Pages custom domain                              |
| <code>base_path</code> | unset   | URL prefix for project Pages (writes <code>&lt;base href="/prefix/"&gt;</code> )                         |
| <code>links</code>     | unset   | Home-page links; each item needs <code>label</code> plus either <code>url</code> or <code>chapter</code> |

Nothing is inferred from chapter titles. If you want a Downloads link on Home, add it explicitly with `['label' => 'Downloads', 'chapter' => '19-downloads.md']`.

Use `base_path` when the site is not at the domain root (for example `https://user.github.io/my-repo/`). Leave it unset when using a custom domain via `cname`.

## Hosting this handbook

In the Papyrus repository, pushes to **main** rebuild the handbook site and publish it to **gh-pages** via **.github/workflows/pages.yml**. Locally:

```
composer build:handbook
# or just:
papyrus build:site -d examples/the-papyrus-handbook -e docs
```

## Mermaid on the site

With `mermaid.theme => auto`, each diagram embeds light and dark SVG variants; CSS swaps them with `data-theme`. Diagrams stay within the reading column (same max-width ladder as the HTML theme).

# build:sample

Build a **sample** PDF for marketing downloads or review copies. Select content with page ranges, whole chapters by filename, or both.

```
papyrus build:sample
papyrus build:sample --theme light
papyrus build:sample -t light,dark
```

**papyrus build** does **not** run this by default — pass **--with-sample**, or call **build:sample** directly.

## Options

| Option                | Short           | Default               | Meaning                     |
|-----------------------|-----------------|-----------------------|-----------------------------|
| <code>--dir</code>    | <code>-d</code> | current directory     | Book root                   |
| <code>--export</code> | <code>-e</code> | <code>export/</code>  | Output directory            |
| <code>--theme</code>  | <code>-t</code> | all configured themes | Comma-separated theme names |

# Config

```
'sample' => [
  'ranges' => [
    ['from' => 1, 'to' => 4],
    ['from' => 10, 'to' => 12],
  ],
  'chapters' => [
    '01-introduction.md',
    '04-writing-content', // stem or basename also works
  ],
  // Optional notice text (also accepted as sample.notice /
  sample.text):
  // 'notice' => 'This is a sample...',
],

'sample_notice' => 'This is a sample from My Book.',
```

| Key                          | Meaning                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sample.ranges</code>   | List of <code>{from, to}</code> <b>1-based inclusive</b> page ranges against the finished PDF                                                                                                         |
| <code>sample.chapters</code> | Chapter files to include in full (config order). Match by source path, basename ( <b>01-intro.md</b> ), or stem ( <b>01-intro</b> ). Rendered as body pages only — no extra cover, title page, or TOC |



| `sample_notice` | Extra notice page (preferred) | | `sample.notice` /  
`sample.text` | Fallbacks if `sample_notice` is empty |

Invalid ranges are dropped. Unknown chapter names fail the build.  
`build:sample` requires at least one of `ranges` or `chapters`.

When both are set, page-range slices from the full theme PDF come first, then chapter pages (in config order).

## Output

```
export/sample-<slug>-<theme>.pdf
```

When a notice is set, it is appended after the selected content.

# kdp

**papyrus kdp** builds every **enabled** Amazon KDP artifact in one shot: Kindle EPUB (if enabled), print interior PDF (if enabled), cover asset copies, and the metadata JSON sidecar.

```
papyrus kdp
papyrus kdp --require-epubcheck
papyrus kdp --package
papyrus kdp --wrap
papyrus kdp -d /path/to/book -e /path/to/out
```

Unlike **papyrus build**, this command **fails** when neither **kdp.ebook.enabled** nor **kdp.print.enabled** is true.

## Options

| Option                           | Short           | Default              | Meaning                                                     |
|----------------------------------|-----------------|----------------------|-------------------------------------------------------------|
| <code>--dir</code>               | <code>-d</code> | current directory    | Book root                                                   |
| <code>--export</code>            | <code>-e</code> | <code>export/</code> | Output directory                                            |
| <code>--require-epubcheck</code> |                 | off                  | Fail ebook build if <code>epubcheck</code> is missing       |
| <code>--package</code>           |                 | off                  | Also write <code>export/&lt;slug&gt;-kdp-package.zip</code> |
| <code>--wrap</code>              |                 | off                  | Also generate wraparound cover PDF + PNG                    |

Print uses the **first** configured theme (same as `kdp:print` without `--theme`).

## Enable flags

```
'kdp' => [
  'ebook' => [
    'enabled' => true,
    'cover' => 'cover-ebook.jpg',
  ],
  'print' => [
    'enabled' => true,
    'bleed_mm' => 3,
    'margin_preset' => 'recommended', // or minimum
    'paper' => 'white',                // metadata + spine
    formula (e.g. cream)
    'back_cover' => 'back-cover.png', // optional wrap
    artwork
    'spine_color' => '#1a1a1a',        // optional wrap spine
    fill
  ],
  'metadata' => [
    'description' => 'Bookstore blurb...',
    'keywords' => ['papyrus', 'markdown'],
    'language' => 'en',
  ],
],
```

| Key                        | Default            | Notes            |
|----------------------------|--------------------|------------------|
| <code>ebook.enabled</code> | <code>false</code> | Gates ebook EPUB |

| Key                              | Default              | Notes                                                  |
|----------------------------------|----------------------|--------------------------------------------------------|
| <code>ebook.cover</code>         | unset                | Asset under <code>assets/</code>                       |
| <code>print.enabled</code>       | <code>false</code>   | Gates print PDF                                        |
| <code>print.bleed_mm</code>      | 3                    | Added to trim and each margin                          |
| <code>print.margin_preset</code> | recommended          | recommended or minimum                                 |
| <code>print.paper</code>         | white                | Spine formula + metadata ( <code>cream</code> allowed) |
| <code>print.back_cover</code>    | unset                | Optional back panel for <code>--wrap</code>            |
| <code>print.spine_color</code>   | sample front cover   | Hex fill for spine / generated back                    |
| <code>metadata.*</code>          | see metadata chapter | Sidcar + EPUB description fallback                     |

## Typical artifacts

| Artifact           | Filename                                                                                                |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Kindle EPUB        | <code>&lt;slug&gt;-kdp.epub</code>                                                                      |
| Print PDF          | <code>&lt;slug&gt;-kdp-print.pdf</code>                                                                 |
| Ebook cover copy   | <code>&lt;slug&gt;-kdp-ebook-cover.&lt;ext&gt;</code>                                                   |
| Print cover copies | <code>&lt;slug&gt;-kdp-print-cover-&lt;theme&gt;.&lt;ext&gt;</code>                                     |
| Wrap cover         | <code>&lt;slug&gt;-kdp-print-wrap.pdf</code> (+ <code>.png</code> preview via <code>-wrap</code> )      |
| Metadata           | <code>&lt;slug&gt;-kdp-metadata.json</code>                                                             |
| Package zip        | <code>&lt;slug&gt;-kdp-package.zip</code> (via <code>kdp:package</code> or <code>kdp --package</code> ) |

After `kdp:print`, Papyrus prints a wrap-cover size estimate (spine + full bleed dimensions). Confirm trim with `papyrus sizes` and Amazon's current KDP print specs before uploading.

## Package for upload

After the artifacts exist:

```
papyrus kdp:package  
# or fold it into the all-in-one run:  
papyrus kdp --package
```

That writes `export/<slug>-kdp-package.zip` containing the enabled ebook / print / cover / metadata files plus a `KDP-CHECKLIST.txt` reminder. It does not talk to Amazon — upload the zip contents through KDP's web UI.

The next chapters document each `kdp:*` command.



# kdp:ebook

Build a Kindle-oriented EPUB for Amazon KDP upload.

```
papyrus kdp:ebook  
papyrus kdp:ebook --require-epubcheck  
papyrus kdp:ebook -d /path/to/book -e /path/to/out
```

Requires `kdp.ebook.enabled => true`.

## Options

| Option                           | Short           | Default              | Meaning                                                    |
|----------------------------------|-----------------|----------------------|------------------------------------------------------------|
| <code>--dir</code>               | <code>-d</code> | current directory    | Book root                                                  |
| <code>--export</code>            | <code>-e</code> | <code>export/</code> | Output directory                                           |
| <code>--require-epubcheck</code> |                 | off                  | Fail if <code>epubcheck</code> is not on <code>PATH</code> |

## Output

```
export/<slug>-kdp.epub
```

Papyrus runs an internal KDP-oriented check and, when available on **PATH**, **epubcheck** (install with **brew install epubcheck**, or from [w3c/epubcheck](https://github.com/w3c/epubcheck)). Warnings from those checks are printed to the console. Without epubcheck, the build still succeeds with a warning unless you pass **-require-epubcheck**.

## Cover

```
'kdp' => [  
  'ebook' => [  
    'enabled' => true,  
    'cover' => 'cover-ebook.jpg', // under assets/  
  ],  
],
```

If `kdp.ebook.cover` is unset, Papyrus falls back to the light theme cover (`cover.light` or `cover.image`).

## Metadata

Description falls back to "{title} - {author}" when `kdp.metadata.description` is empty. Language uses `kdp.metadata.language` when set, otherwise project `language` (default `en`).

## Versus build:epub

|                     | <code>build:epub</code>        | <code>kdp:ebook</code>             |
|---------------------|--------------------------------|------------------------------------|
| Filename            | <code>&lt;slug&gt;.epub</code> | <code>&lt;slug&gt;-kdp.epub</code> |
| Enable flag         | always                         | <code>kdp.ebook.enabled</code>     |
| Validation          | packaging only                 | KDP checks + optional<br>epubcheck |
| Cover<br>preference | theme cover                    | <code>kdp.ebook.cover</code> first |

Internal checks also warn on an empty `kdp.metadata.description` and on eBook cover images smaller than 1600 px on the shortest side (when GD can read the file).

# kdp:print

Build a **print interior** PDF sized for Amazon KDP: no cover page, KDP margin presets, and bleed applied around the project trim.

```
papyrus kdp:print  
papyrus kdp:print --theme light  
papyrus kdp:print -t dark -e /path/to/out
```

Requires `kdp.print.enabled => true`.

## Options

| Option                | Short           | Default                | Meaning                       |
|-----------------------|-----------------|------------------------|-------------------------------|
| <code>--dir</code>    | <code>-d</code> | current directory      | Book root                     |
| <code>--export</code> | <code>-e</code> | <code>export/</code>   | Output directory              |
| <code>--theme</code>  | <code>-t</code> | first configured theme | PDF theme for interior layout |



## Output

```
export/<slug>-kdp-print.pdf
```

## Print config

```
'kdp' => [
  'print' => [
    'enabled' => true,
    'bleed_mm' => 3,
    'margin_preset' => 'recommended', // or 'minimum'
    'paper' => 'white',               // cream, etc. -
  ],
  metadata only
],
```

## Bleed

Bleed expands the page and margins:

- Page width / height each increase by  $2 \times \text{bleed\_mm}$
- Every margin increases by  $\text{bleed\_mm}$

Default `bleed_mm` is 3.

## Margin presets (mm, before bleed)

| Preset      | Left | Right | Top  | Bottom |
|-------------|------|-------|------|--------|
| recommended | 12.7 | 9.525 | 12.7 | 12.7   |

| Preset               | Left  | Right | Top   | Bottom |
|----------------------|-------|-------|-------|--------|
| <code>minimum</code> | 9.525 | 6.35  | 9.525 | 9.525  |

Unknown preset names fall back to `recommended`.

## Paper

`print.paper` is written into the KDP metadata sidecar and used for wrap-cover spine estimates (`white` or `cream`). It does not change PDF colour or stock — choose white vs cream in the KDP dashboard to match.

After a successful build, Papyrus prints a wrap-cover estimate from the interior page count (Amazon's published inches-per-page formula + 0.125" bleed). Use `kdp:cover --dimensions` to recalculate later.

## Trim size

Uses the same `document.size` / `document.format` as your regular PDFs. `papyrus doctor` warns when trim falls outside KDP bounds (width 101.6–215.9 mm, height 152.4–279.4 mm). List presets with `papyrus sizes`.

## Cover

The print PDF **omits** the cover page. Export wraparound / ebook cover files with **kdp:cover**.

# kdp:cover

Copy configured cover images from **assets/** into **export/** with KDP-oriented filenames. No enable flag — safe to run anytime.

```
papyrus kdp:cover
papyrus kdp:cover -d /path/to/book -e /path/to/out
papyrus kdp:cover --dimensions --pages=220
papyrus kdp:cover --wrap --pages=220
papyrus kdp:cover --wrap --theme=light
```

## Options

| Option                    | Short           | Default              | Meaning                                                        |
|---------------------------|-----------------|----------------------|----------------------------------------------------------------|
| <code>--dir</code>        | <code>-d</code> | current directory    | Book root                                                      |
| <code>--export</code>     | <code>-e</code> | <code>export/</code> | Output directory                                               |
| <code>--dimensions</code> |                 | off                  | Print wrap-cover size estimate (spine + full bleed)            |
| <code>--wrap</code>       |                 | off                  | Generate wraparound cover PDF + PNG preview                    |
| <code>--pages</code>      |                 | from print PDF       | Page count for <code>--dimensions</code> / <code>--wrap</code> |
| <code>--theme</code>      |                 | first theme          | Front-cover theme used by <code>--wrap</code>                  |

## Sources and outputs

| Source                                                                | Export name                                                                            |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>kdp.ebook.cover</code><br>→<br><code>assets/&lt;file&gt;</code> | <code>&lt;slug&gt;-kdp-ebook-cover.&lt;ext&gt;</code>                                  |
| <code>cover[&lt;theme&gt;]</code><br>or <code>cover.image</code>      | <code>&lt;slug&gt;-kdp-print-cover-&lt;theme&gt;.&lt;ext&gt;</code><br>(one per theme) |

Missing source files are skipped (no error). If nothing is copied, the command still succeeds with a comment.



## Example

```
'cover' => [  
  'image' => 'cover.png',  
  'light' => 'cover-light.png',  
],  
'kdp' => [  
  'ebook' => [  
    'cover' => 'cover-ebook.jpg',  
  ],  
],
```

With themes **light** and **dark**, a typical export folder might contain:

```
export/my-book-kdp-ebook-cover.jpg  
export/my-book-kdp-print-cover-light.png  
export/my-book-kdp-print-cover-dark.png    # if dark cover  
resolves
```

Papyrus copies files as-is for ebook/front covers. With **--wrap** it also composes a **paperback wraparound** PDF (back | spine | front, plus bleed) using Amazon's spine formula and your front cover image:

```
papyrus kdp:cover --wrap  
papyrus kdp:cover --wrap --pages=220
```

Outputs:

```
export/<slug>-kdp-print-wrap.pdf # upload to KDP as the cover
export/<slug>-kdp-print-wrap.png # preview
```

Optional config:

```
'cover' => [
  'image' => 'cover.png',
  'back' => 'back-cover.png', // optional; otherwise a text
back panel is drawn
],
'kdp' => [
  'print' => [
    'paper' => 'white', // or cream - affects spine
width
    'back_cover' => 'back-cover.png', // preferred over
cover.back
    'spine_color' => '#1a1a1a', // optional; default
samples the front cover
  ],
],
```

Without `--pages`, Papyrus counts pages in `export/<slug>-kdp-print.pdf` (build `kdp:print` first). Spine text is drawn only when Amazon allows it ( $\geq 79$  pages). Always verify with Amazon's cover calculator before upload.

# kdp:metadata

Emit a JSON sidecar you can keep next to KDP uploads or feed into your own publishing scripts. Always writes — no enable flag.

```
papyrus kdp:metadata  
papyrus kdp:metadata -d /path/to/book -e /path/to/out
```

## Options

| Option                | Short           | Default              | Meaning          |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code>    | <code>-d</code> | current directory    | Book root        |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

## Output

```
export/<slug>-kdp-metadata.json
```

## Config

```
'language' => 'en',

'kdp' => [
  'print' => [
    'paper' => 'white',
    'bleed_mm' => 3,
    'margin_preset' => 'recommended',
  ],
  'metadata' => [
    'description' => 'Bookstore blurb...',
    'keywords' => ['papyrus', 'markdown', 'ebook'],
    'language' => 'en', // optional; else project language
  ],
],
```

## JSON shape

```
{
  "title": "...",
  "subtitle": "...",
  "author": "...",
  "language": "en",
  "description": "...",
  "keywords": [...],
  "ebook": {
    "enabled": true,
    "cover": "cover-ebook.jpg",
    "artifact": "<slug>-kdp.epub"
  },
  "print": {
    "enabled": true,
    "paper": "white",
    "bleed_mm": 3,
    "margin_preset": "recommended",
    "margin_preset_known": true,
    "trim": {
      "width_mm": 152.4,
      "height_mm": 228.6,
      "width_in": 6.0,
      "height_in": 9.0,
      "preset": "6x9",
      "within_kdp_bounds": true
    },
    "artifact": "<slug>-kdp-print.pdf"
  },
  "artifacts": {
    "ebook": "<slug>-kdp.epub",
```

```

    "print": "<slug>-kdp-print.pdf",
    "metadata": "<slug>-kdp-metadata.json",
    "ebook_cover": "<slug>-kdp-ebook-cover.jpg",
    "print_covers": ["<slug>-kdp-print-cover-light.jpg"]
  }
}

```

| Field                                  | Source                                                                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>title / subtitle / author</code> | Project identity                                                                                                      |
| <code>language</code>                  | <code>kdp.metadata.language</code> → <code>language</code> → <code>en</code>                                          |
| <code>description</code>               | <code>kdp.metadata.description</code> (may be empty)                                                                  |
| <code>keywords</code>                  | <code>kdp.metadata.keywords</code> (string list)                                                                      |
| <code>ebook.* / print.*</code>         | Enable flags, paper, bleed, margin preset, trim size; optional wrap <code>cover</code> estimate when print PDF exists |
| <code>artifacts.*</code>               | Expected export filenames for uploads                                                                                 |

When `export/<slug>-kdp-print.pdf` exists, `print.cover` includes page count, spine width, and full wrap dimensions. Rebuild metadata after `kdp:print` to refresh those numbers.

This file is a helper for your workflow — it is not uploaded automatically to



Amazon. Run **papyrus doctor** to check KDP readiness (covers, margin preset, epubcheck on **PATH**). Package uploads with **papyrus kdp:package**.

# Migration and CI

## Migrating from ibis-next

```
papyrus migrate-ibis
papyrus migrate-ibis -d /path/to/book
papyrus migrate-ibis --force
```

| Option               | Short           | Meaning                                        |
|----------------------|-----------------|------------------------------------------------|
| <code>--dir</code>   | <code>-d</code> | Book root                                      |
| <code>--force</code> | <code>-f</code> | Overwrite an existing <code>papyrus.php</code> |

This writes `papyrus.php` from `ibis.php`. If you still have custom `assets/theme*.html` files from ibis, it rewrites `<!-- IBIS:TOC -->` to `<!-- PAPYRUS:TOC -->` in those **project** files only. Bundled Papyrus themes (used when `assets/` has no theme override) already use the Papyrus marker — `migrate-ibis` does not copy or rewrite vendor assets.

After migration:

1. Remove `hi-folks/ibis-next` and any Composer patches for `ibis/mPDF`.
2. Point scripts at `papyrus` (`build`, `build:pdf`, `build:site`, ...).
3. Decide on themes: keep your ibis HTML under `assets/` as overrides, or delete those copies and use Papyrus defaults (optional `papyrus asset:publish` later if you want local copies to edit).
4. Run `papyrus doctor` and a full `papyrus build`.

## Continuous integration

Copy the stub workflow from the Papyrus package:

```
stubs/github/workflows/book-build.yml
```

A typical book job installs PHP with `dom`, `gd`, `mbstring`, `zip`, `zlib`, runs `composer install`, `papyrus doctor`, optional `papyrus lint`, then `composer build`.

To publish a `build:site` output on GitHub Pages, add a job that runs `papyrus build:site` and deploys the site directory to a `gh-pages` branch (see [pages.yml](#) in this repository for a working example, including Chromium for Mermaid).

## Programmatic use

```
use Milon\Papyrus\Config\Project;

$project = Project::load($bookDir);
$book =
$project->bookConverter()->convertDirectory($project->contentDir)
;
```

## Checklist before release

1. `papyrus doctor` is clean
2. `papyrus build` succeeds for every theme
3. `papyrus build:site` looks right on phone and desktop (try light and dark)
4. Sample ranges still make sense after reflows (`build:sample`)
5. KDP EPUB opens in Kindle Previewer; print PDF matches trim/bleed
6. Pin a Papyrus version in the book's `composer.json`

Report problems at [github.com/milon/papyrus/issues](https://github.com/milon/papyrus/issues).

# Command reference

# Build

| Command                   | Purpose                                                                        | Notable options                                                   |
|---------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <code>build</code>        | PDF themes, EPUB, HTML, enabled KDP                                            | <code>-d, -e, --with-site, --with-sample, --include-drafts</code> |
| <code>build:pdf</code>    | PDF themes                                                                     | <code>--theme, --parallel</code>                                  |
| <code>build:epub</code>   | EPUB3                                                                          | <code>-d, -e</code>                                               |
| <code>build:html</code>   | Single-file HTML                                                               | <code>-d, -e</code>                                               |
| <code>build:site</code>   | Multi-page HTML site                                                           | <code>-d, -e</code>                                               |
| <code>build:sample</code> | Sample PDF from <code>sample.ranges</code> and/or <code>sample.chapters</code> | <code>--theme</code>                                              |



## KDP

| Command                   | Purpose                       | Notable options                                                                                                     |
|---------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>kdp</code>          | All enabled KDP outputs       | <code>-d</code> , <code>-e</code> , <code>--require-epubcheck</code> , <code>--package</code> , <code>--wrap</code> |
| <code>kdp:ebook</code>    | Kindle EPUB                   | <code>--require-epubcheck</code> ; requires <code>kdp.ebook.enabled</code>                                          |
| <code>kdp:print</code>    | Print interior PDF            | <code>--theme</code> ; requires <code>kdp.print.enabled</code>                                                      |
| <code>kdp:cover</code>    | Cover assets + optional wrap  | <code>--dimensions</code> , <code>--wrap</code> , <code>--pages</code> , <code>--theme</code>                       |
| <code>kdp:metadata</code> | Metadata JSON sidecar         | no enable flag                                                                                                      |
| <code>kdp:package</code>  | Zip KDP artifacts + checklist | requires enabled outputs + built files                                                                              |

## Project tooling

| Command                    | Purpose                                                                                                      | Notable options                                                                                                 |
|----------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>init</code>          | Scaffold <code>papyrus.php</code> , <code>content/</code> , and an empty <code>assets/</code>                | <code>--force</code>                                                                                            |
| <code>asset:publish</code> | Publish bundled themes, CSS, and fonts into <code>assets/</code>                                             | <code>--force</code> , <code>--only=themes,css,fonts</code>                                                     |
| <code>doctor</code>        | Validate config, paths, Mermaid, KDP readiness                                                               | <code>-d</code>                                                                                                 |
| <code>sizes</code>         | List page-size presets (+ KDP in-bounds)                                                                     | (no <code>-d</code> / <code>-e</code> )                                                                         |
| <code>lint</code>          | Lint PHP fences in <code>content/</code>                                                                     | <code>--fix</code> , <code>--max-width=66</code>                                                                |
| <code>watch</code>         | Rebuild via <code>build</code> on file changes                                                               | <code>--interval</code> , <code>--with-site</code> , <code>--with-sample</code> , <code>--include-drafts</code> |
| <code>serve</code>         | Preview the generated site in a browser                                                                      | <code>--host</code> , <code>-p/--port</code> , <code>--build</code> , <code>-s/--site</code>                    |
| <code>migrate-ibis</code>  | <code>ibis.php</code> → <code>papyrus.php</code> ; TOC markers in local <code>assets/theme*.html</code> only | <code>--force</code>                                                                                            |



## Shared book options

Most book commands accept:

| Option                        | Short           | Default                          | Meaning                                              |
|-------------------------------|-----------------|----------------------------------|------------------------------------------------------|
| <code>--dir</code>            | <code>-d</code> | current directory                | Book root<br>(contains <code>papyrus.php</code> )    |
| <code>--export</code>         | <code>-e</code> | <code>&lt;book&gt;/export</code> | Override<br>export<br>directory                      |
| <code>--include-drafts</code> |                 | off                              | Include<br>chapters with<br><code>draft: true</code> |

Also useful globally: `-v` / `--verbose` (include third-party vendor notices), `-q` / `--quiet`, `--no-ansi`.

## Output filenames

| Command                    | Path under export                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------|
| <code>build:pdf</code>     | <code>&lt;slug&gt;-&lt;theme&gt;.pdf</code>                                               |
| <code>build:html</code>    | <code>&lt;slug&gt;.html</code>                                                            |
| <code>build:epub</code>    | <code>&lt;slug&gt;.epub</code>                                                            |
| <code>build:site</code>    | <code>&lt;slug&gt;-site/</code>                                                           |
| <code>build:sample</code>  | <code>sample-&lt;slug&gt;-&lt;theme&gt;.pdf</code>                                        |
| <code>kdp:ebook</code>     | <code>&lt;slug&gt;-kdp.epub</code>                                                        |
| <code>kdp:print</code>     | <code>&lt;slug&gt;-kdp-print.pdf</code>                                                   |
| <code>kdp:cover</code>     | <code>&lt;slug&gt;-kdp-ebook-cover.*, &lt;slug&gt;-kdp-print-cover-&lt;theme&gt;.*</code> |
| <code>wrap (--wrap)</code> | <code>&lt;slug&gt;-kdp-print-wrap.pdf, &lt;slug&gt;-kdp-print-wrap.png</code>             |
| <code>kdp:metadata</code>  | <code>&lt;slug&gt;-kdp-metadata.json</code>                                               |
| <code>kdp:package</code>   | <code>&lt;slug&gt;-kdp-package.zip</code>                                                 |

Packagist: [milon/papyrus](https://packagist.org/packages/milon/papyrus). Source: [github.com/milon/papyrus](https://github.com/milon/papyrus). Releases: [github.com/milon/papyrus/releases](https://github.com/milon/papyrus/releases).

# Downloads

Preview this handbook without building locally. Files live in the Papyrus repository under [docs/](#) and update when the handbook is rebuilt.

## Full PDF

| Theme | Download                                              |
|-------|-------------------------------------------------------|
| Light | <a href="#"><u>the-papyrus-handbook-light.pdf</u></a> |
| Dark  | <a href="#"><u>the-papyrus-handbook-dark.pdf</u></a>  |

On GitHub: [light](#) · [dark](#)

## Sample PDF

A shorter preview built with `build:sample`: cover through Welcome (before the table of contents), then the `build:html` and `kdp:cover` chapters.

| Theme | Download                                                     |
|-------|--------------------------------------------------------------|
| Light | <a href="#"><u>sample-the-papyrus-handbook-light.pdf</u></a> |
| Dark  | <a href="#"><u>sample-the-papyrus-handbook-dark.pdf</u></a>  |

Configured in this book's `papyrus.php` as:

```
'sample' => [  
  'ranges' => [  
    ['from' => 1, 'to' => 5], // cover ... Welcome (before TOC)  
  ],  
  'chapters' => [  
    '09-build-html.md',  
    '15-kdp-cover.md',  
  ],  
],
```



## Other previews

| Format           | Link                                                                      |
|------------------|---------------------------------------------------------------------------|
| This site        | <a href="https://papyrus.milon.im">papyrus.milon.im</a>                   |
| Single-file HTML | <a href="https://the-papyrus-handbook.html">the-papyrus-handbook.html</a> |

Rebuild everything from the Papyrus repo with:

```
composer build:handbook
```